BANK LOAN APPROVAL PREDICTION USING LOGISTIC REGRESSION

Github: Project Repository

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PROBLEM AND DATA:

- **Size: 5000 rows x 14 columns**

- Dataset: Bank Loan Approval dataset (Kaggle): LINK

Banks need to decide which customers are likely to be approved for a personal loan. We predict whether a customer's loan is approved (1) or not (0) from details such as income, age, credit card spending and mortgage amount. This can make lending decisions faster and more consistent.

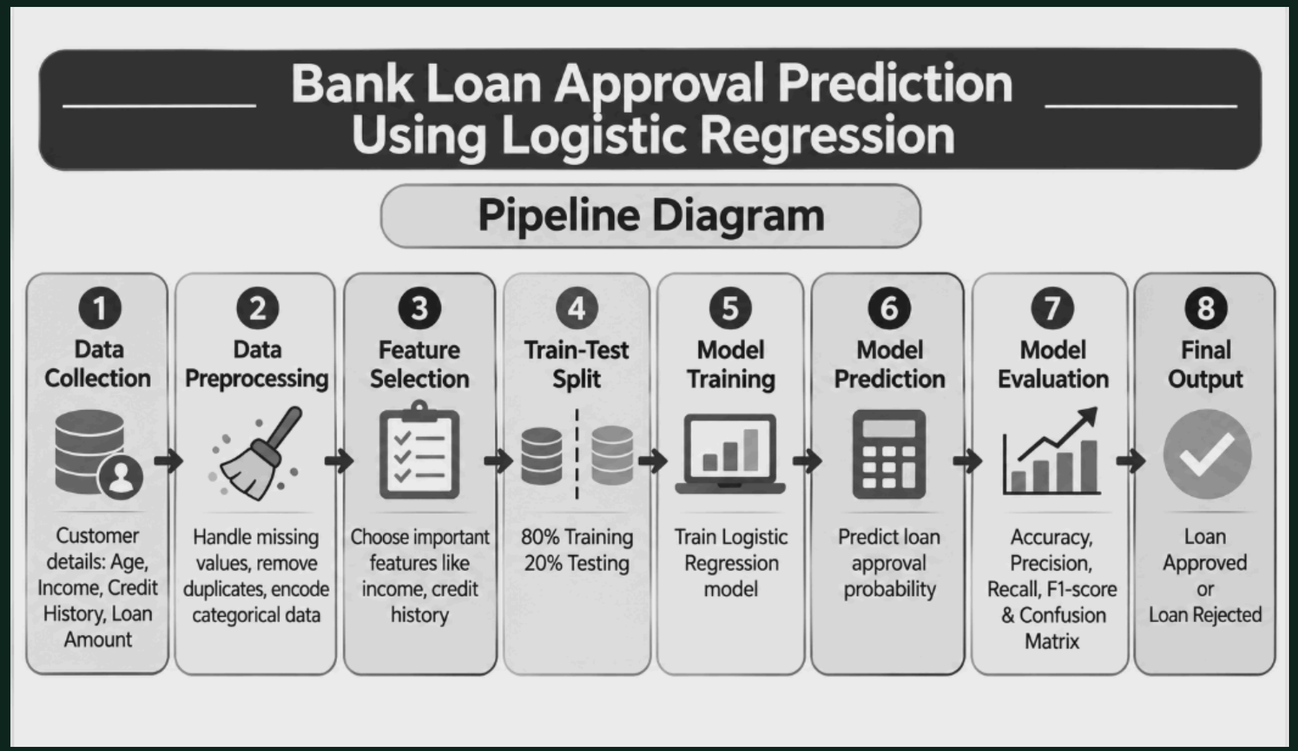
Target variable:

- Personal Loan (Binary: 1 for approved/accepted, 0 for rejected/not accepted)

Mission

Deliver secure, innovative, and customer-focused financial solutions that create long-term value

APPROCH USED – LINEAR REGRESSION



- Logistic Regression is a supervised Machine Learning algorithm used to predict categorical binary outcomes (such as 0 or 1, Yes or No).
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- In our project, it predicts personal loan approval status based on different input factors such as credit score, annual income, loan amount, employment history, and other financial details.

Logistic Regression Formula:

$$P(Y = 1) = 1 / (1 + e^{-(b^0+b^1x^1+b^2x^2+\ldots+bnxn)})$$

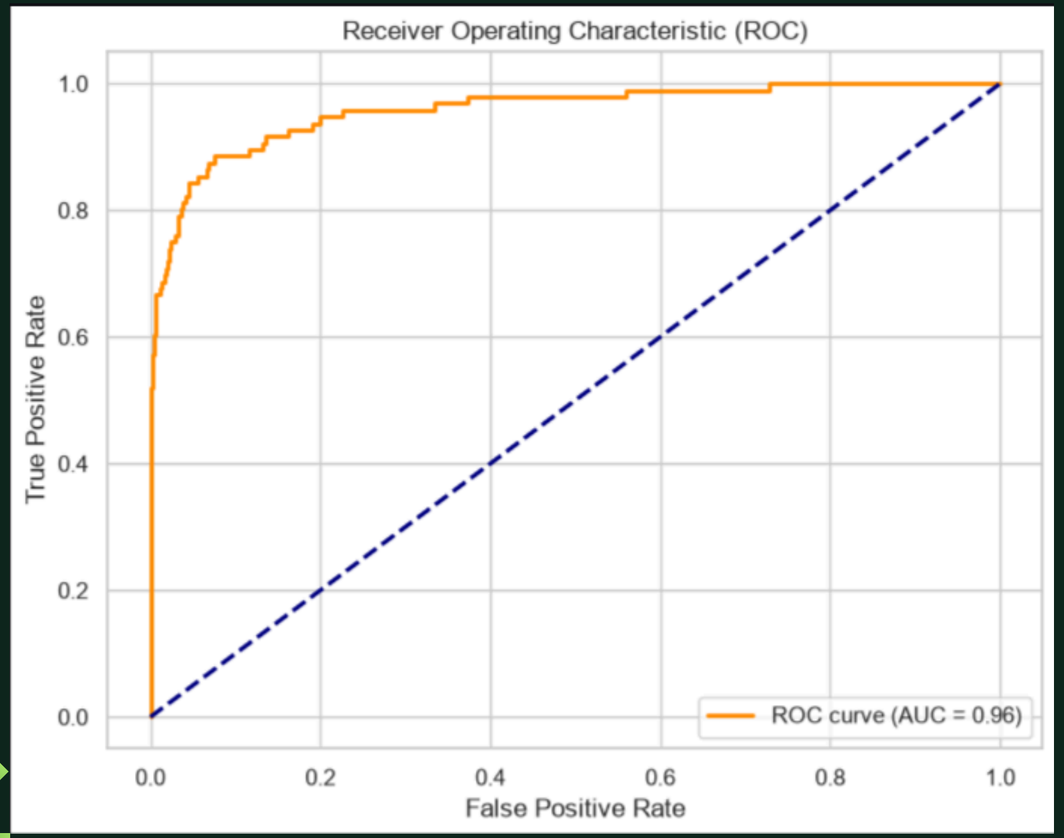
Where:

P(Y = 1) = Probability of personal loan approval.
b_0 = Intercept.

b_1, b_2, ……., b_n = Model coefficients.
X_1, X_2, ……., X_n = Input features.

RESULT OF THE PROJECT:

- After training and testing the Logistic Regression model, we obtain predicted loan approval outcomes based on the given applicant information.



- Accuracy Score:
  - Measures the overall correctness of the model's classifications. A value closer to 1 generally indicates a better fit.
- Precision:
  - Measures the proportion of correctly predicted positive loan approvals out of all predicted positive outcomes.
- Recall :
  - Measures the proportion of actual approved loans correctly identified by the model.

LEARNINGS

1.What Worked?

➜Logistic Regression effectively predicts binary outcome states for loan approvals (Approved vs. Rejected).

➜Matplotlib and Seaborn helped visualize confusion matrices, ROC-AUC curves, and key feature correlations.

2.What Didn't Work?

➜Classification errors occurred on edge cases, particularly for applicants with borderline credit scores or non-traditional income.

➜Logistic Regression struggled to capture non-linear relationships.Class imbalance skewed initial predictions toward the majority class.

3.AI- Use Disclosure

➜AI tools were used to support domain context understanding, assist with Python code optimization, and assist in slide deck presentation preparation. All dataset preprocessing, model training, evaluation, and results synthesis were conducted directly as part of our project implementation.

THANK YOU!